Library
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Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Computer Science/ Computer Systems Engineering

Repeat Examination
Year 2, Semester 1 (2026)

SE2032 – Database Management Systems

Duration: 3 Hours

May 2026

Instructions to Candidates:

- This paper has 5 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 6 pages, including the cover page.
- Scientific calculators are allowed. Any other devices capable of storing and retrieving text, including mobile phones are not allowed.
- Removing this examination paper from this examination hall or sharing any part of its content is considered an Examination offence.

Question 1

(20 Marks)

A database designer is designing a database for a Logistics Management System. The following requirements have been gathered:

At a logistics company, managers coordinate delivery operations involving drivers and shipment routes. Each manager has a unique ManagerID, Name, Email, and Department. Managers supervise delivery routes. The route has a unique RouteID, Destination, StartDate, and EstimatedCost. Each route must be supervised by exactly one manager, but a manager may supervise multiple routes.

Every driver has a DriverID, Name, Email, and LicenseType. A driver can participate in multiple routes, and in each route, they are assigned a specific responsibility (e.g., Main Driver, Backup Driver). Additionally, drivers may mentor other drivers as part of a training program. Drivers can be further categorized into FullTimeDrivers and ContractDrivers with attributes specific to their category: FullTimeDrivers may have MonthlySalary and JoiningDate, while ContractDrivers may have ContractDuration.

The delivery system includes shipment orders with attributes such as OrderNumber, Weight, DeliveryPriority, and DeliveryDate. The company records which manager approved each shipment order for a particular route with ApprovalDate indicating when the manager authorized the shipment.

For every route, a number of InspectionReports are prepared to record vehicle and delivery status details. Each InspectionReport depends on a single Route. It is identified by a ReportNumber that is unique only within the context of its Route. The InspectionReport includes ReportNumber, InspectionType, Remarks, and InspectionDate.Construct an EER Diagram that includes:

(a) All relevant entities and their attributes. **(6 marks)**

(b) Appropriate relationships, including recursive (if any), binary, ternary (if any) and superclass/ subclass (ISA) relationships. **(6 marks)**

(c) Clear participation constraints and cardinalities. **(4 marks)**

(d) Proper indication of primary keys, partial keys, and descriptive attributes. Note: State any assumptions that you make. **(4 marks)**

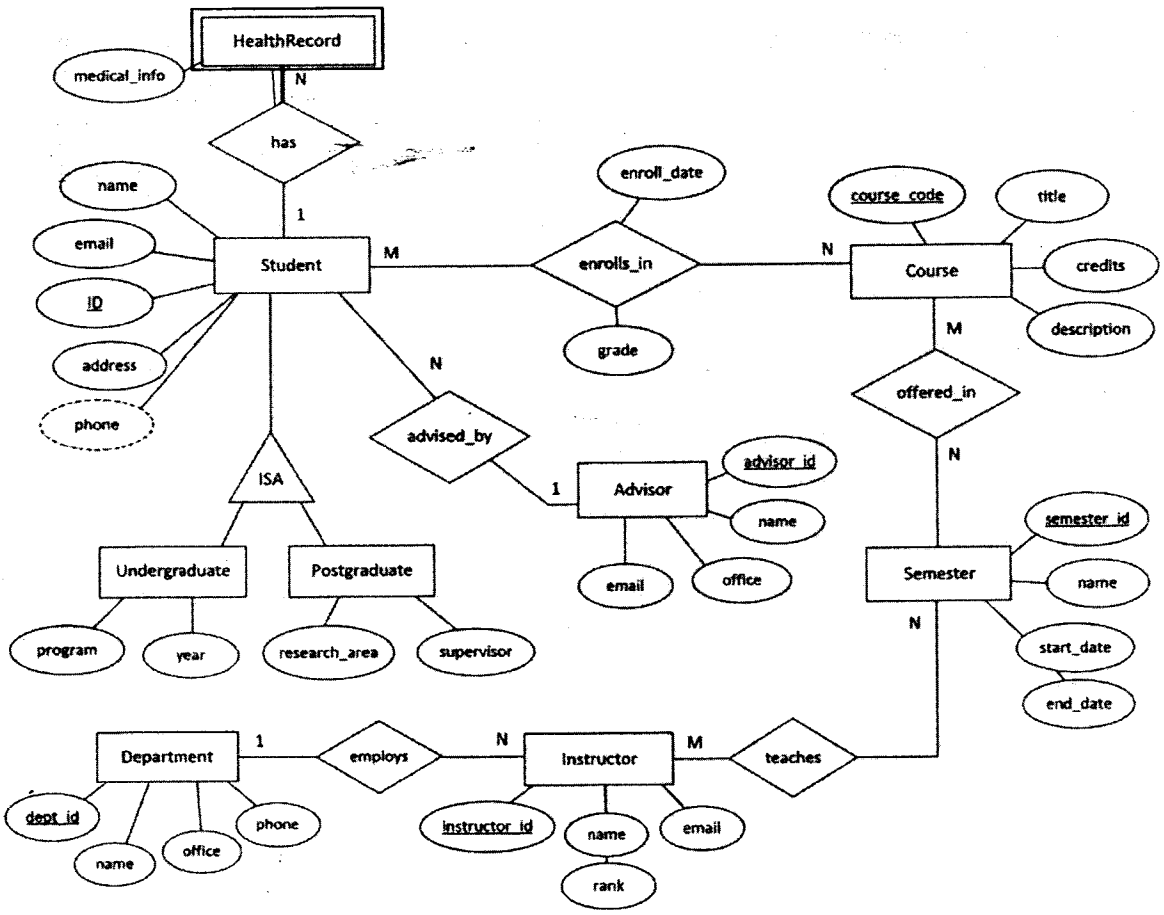
Question 2 (20 marks)

(a) Consider the following EER model.

i. What is the best option for mapping the ISA hierarchy in the diagram? Justify your answer.

(3 Marks)

ii. Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulted relations clearly. **(10 Marks)**



Undergraduate and Postgraduate cover Student

(b) Consider the following relation and functional dependencies FD1, FD2, and FD3.

EmployeeProject (EmployeeID, ProjectID, ProjectManager, Department, HoursWorked)

FD1: EmployeeID, ProjectID → HoursWorked

FD2: ProjectID → ProjectManager

FD3: ProjectManager → Department

i. Find all the keys using attribute closure method. **(2 Marks)**

ii. Is relation in BCNF? Give reasons for your conclusion. If relation is not in BCNF, convert it to a set of BCNF relations. **(5 Marks)**

Question 3 (20 Marks)

(a) Explain the Atomicity property with an example. **(3 Marks)**

(b) Consider the following schedules S1 and S2.

S1: R1(X), W1(X), R2(X), W2(X)
S2: R1(Y), R2(Y), W1(Y), W2(Y), R1(Z)

Here, Ri(X) means read the item X of transaction i, Wi(X) means write item X of transaction i.

i. By drawing a precedence graph, identify the schedule that is serializable and that is not serializable. Justify your answer. **(4 Marks)**

ii. Give the possible serial schedule for the identified serializable schedule. **(2 Marks)**

(c) Consider the following interleaved execution involving transactions T1 and T2, where both access and modify data items S1_Seats and BookingCount.

Transaction T1 reserves 5 seats from a movie theater booking system for Show S1. Transaction T2 applies a 20% promotional increase to the available seat count of Show S1 after adding temporary bonus seats for a festival event.

T1	T2
Read(S1_Seats)	
	Read(S1_Seats)
	S1_Seats = S1_Seats + (S1_Seats * 0.20)
	Write(S1_Seats)
Read(S1_Seats)	
S1_Seats = S1_Seats - 5	
Write(S1_Seats)	
Read(BookingCount)	
BookingCount = BookingCount + 5	
Write(BookingCount)	

i. Identify and explain the concurrency control problem that occurs in the above schedule. **(2 Marks)**

ii. After applying the Two-Phase Locking (2PL) protocol to prevent the problem identified in (i), an undesirable effect may still occur. Identify this effect and explain how it occurs in this schedule. **(3 Marks)**

iii. Explain how 2PL overcomes the concurrency problem identified in part (i). **(3 Marks)**

iv. Suppose the same schedule is controlled using the Timestamp Ordering (TO) protocol instead of 2PL. Assume the timestamps are assigned as:

TS(T1) = 12 and TS(T2) = 20 (T1 starts earlier).

For each operation in the schedule, determine whether it executes or causes an abort, and explain why. **(3 Marks)**

Question 4 (20 Marks)

Consider the following schema.

Student (SID, SName, City)
Course (CID, CName, Credits)
Lecturer (LID, LName, Department)
Enrollment (SID, CID, LID, Semester)

(a) Express each of the following queries as expressions in relational algebra:

i. Retrieve names of lecturers teaching courses with more than 3 credits. **(2 Marks)**

ii. List all course IDs that have never been enrolled. **(2 Marks)**

iii. Find students enrolled in both Semester 1 and Semester 2. **(2 Marks)**

iv. Retrieve names of students enrolled in all courses. **(2 Marks)**

v. List each course ID with the total number of students enrolled. **(2 Marks)**

(b) Express the above queries (i) to (iv) in tuple relational calculus. **(10 Marks)**

Question 5 (20 Marks)

(a) A file contains 80,000 fixed-length employee records. Record size = 100 bytes, Block size = 500 bytes with unspanned organization.

Disk parameters:

- Seek time = 20 ms
- Block transfer time = 0.8 ms
- Rotation speed = 6000 RPM

Answer the following:

i. Calculate the average rotational delay of the disk. **(2 Marks)**

ii. Calculate the blocking factor (bfr) of the file. **(2 Marks)**

iii. Calculate the number of blocks required to store the file. **(2 Marks)**

iv. Calculate average linear search time if blocks are contiguous. **(2 Marks)**

v. Calculate binary search time assuming file is sorted. **(3 Marks)**

(b) i. You are given two application scenarios. For each one, identify the most appropriate type of database (Document, In Memory, Graph, Spatial, Vector). **(3 Marks)**

A. A recommendation engine compares user preferences and embeddings.

B. A navigation system identifies nearby fuel stations.

C. A gaming application processes live leaderboard updates instantly.

ii. The following table represents a relational model for storing book information in a library system.

Customer

CID	Name	City
C1	Alice	London
C2	Bob	New York
C3	Charlie	Sydney

Purchase

PID	CID	Item	Quantity
P1	C1	Laptop	1
P2	C1	Mouse	2
P3	C2	Smartphone	1
P4	C3	Headphones	3

Convert the data into:

A. Document database format. **(3 Marks)**

B. Graph database representation. **(3 Marks)**

---END OF QUESTION PAPER--